# Discrete Mathematics

## Lecture-2

### Relations

**Ordered pair:** An ordered pair of elements $a$ and $b$, where $a$ is designated as the first element and $b$ as the second element, is denoted by $(a, b)$.

**Product sets:** Consider two arbitrary sets $A$ and $B$. The set of all ordered pairs $(a, b)$ where $a \in A$ and $b \in B$ is called the product, or Cartesian product, of A and B. A short designation of this product is $A \times B$, which is read "$A$ cross $B$." By definition,

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$ One

frequently writes $A^2$ instead of $A \times A$.

**Example.** Let $A = \{1,2\}$, and $B = \{a, b, c\}$. Then

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$$

$$B \times A = \{(a, 1), (b, 1), (c, 1), (a, 2), (b, 2), (c, 2)\}$$

Also, $A \times A = \{(1,1), (1,2), (2,1), (2,2)\}$

Here, $A \times B \neq B \times A$.

The Cartesian product deals with ordered pairs, so naturally the order in which the sets are considered is important. Secondly, using n(S) for the number of elements in a set S, we have:

$$n\,(A \times B) = 6 = 2(3) = n(A)n(B)$$

In fact, $n\,(A \times B) = n(A)n(B)$ for any finite sets $A$ and $B$. This follows from the observation that, for an ordered pair $(a, b)$ in $A \times B$, there are $n(A)$ possibilities for $a$, and for each of these there are $n(B)$ possibilities for $b$.

**Relations:** Let $A$ and $B$ be sets. A binary relation or, simply, relation from $A$ to $B$ is a subset of $A \times B$. Suppose $R$ is a relation from $A$ to $B$. Then $R$ is a set of ordered pairs where each first element comes from $A$ and each second element comes from $B$. That is, for each pair $a \in A$ and $b \in B$, exactly one of the following is true:

(i) $(a, b) \in R$; we then say "$a$ is R-related to $b$", written $aRb$.

(ii) $(a, b) \notin R$; we then say "a is not R-related to b", written $a$

If $R$ is a relation from a set $A$ to itself, that is, if $R$ is a subset of $A^2 = A \times A$, then we say that $R$ is a relation on $A$.

The domain of a relation $R$ is the set of all first elements of the ordered pairs which belong to $R$, and the range is the set of second elements.

**Example.** $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$, and let $R = \{(1, y), (1, z), (3, y)\}$. Then $R$ is a relation from $A$ to $B$ since $R$ is a subset of $A \times B$. With respect to this relation,

$1Ry$, $1Rz$, $3Ry$, but $1\not{R}x$, $2\not{R}x$, $2\not{R}y$, $2\not{R}z$, $3\not{R}x$, $3\not{R}z$

The domain of $R$ is $\{1, 3\}$ and the range is $\{y, z\}$.

## Inverse Relation:

Let $R$ be any relation from a set $A$ to a set $B$. The inverse of $R$, denoted by $R^{-1}$, is the relation from $B$ to $A$ which consists of those ordered pairs which, when reversed, belong to $R$; that is,

$$R^{-1} = \{(b, a) \mid (a, b) \in R\}$$

**For example**, Let $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$. Then the inverse of $R = \{(1, y), (1, z), (3, y)\}$ is $R^{-1} = \{(y, 1), (z, 1), (y, 3)\}$

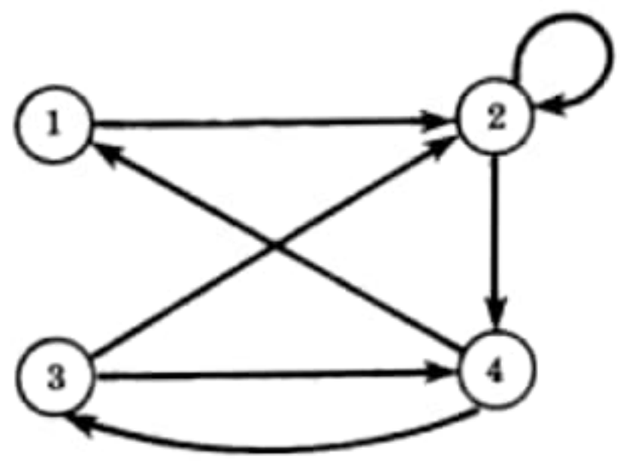
Clearly, if $R$ is any relation, then $(R^{-1})^{-1} = R$. Also, the domain and range of $R^{-1}$ are equal, respectively, to the range and domain of $R$. Moreover, if $R$ is a relation on $A$, then $R^{-1}$ is also a relation on $A$.

## Directed Graphs of Relations on Sets:

There is an important way of picturing a relation $R$ on a finite set. First, we write down the elements of the set, and then we draw an arrow from each element $x$ to each element $y$ whenever $x$ is related to $y$. This diagram is called the directed graph of the relation.

For example, the directed graph of the following relation $R$ on the set $A = \{1, 2, 3, 4\}$:

$$R = \{(1, 2), (2, 2), (2, 4), (3, 2), (3, 4), (4, 1), (4, 3)\} \text{ is}$$




Observe that there is an arrow from 2 to itself, since 2 is related to 2 under $R$.

## Pictures of Relations on Finite Sets:

Suppose $A$ and $B$ are finite sets. There are two ways of picturing a relation $R$ from $A$ to $B$.

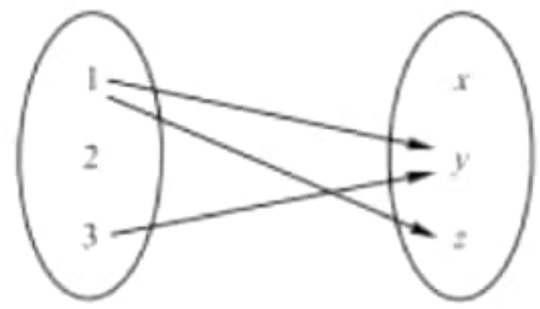
(i) Form a rectangular array (matrix) whose rows are labeled by the elements of $A$ and whose columns are labeled by the elements of $B$. Put a 1 or 0 in each position of the array according as $a \in A$ is or is not related to $b \in B$. This array is called the matrix of the relation.

(ii) Write down the elements of $A$ and the elements of $B$ in two disjoint disks, and then draw an arrow from $a \in A$ to $b \in B$ whenever $a$ is related to $b$. This picture will be called the arrow diagram of the relation.

The following figure pictures the relation $R = \{(1, y), (1, z), (3, y)\}$ (For $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$) by the above two ways.

| | $x$ | $y$ | $z$ |
|---|---|---|---|
| 1 | 0 | 1 | 1 |
| 2 | 0 | 0 | 0 |
| 3 | 0 | 1 | 0 |

(i)




(ii)

$R = \{(1, y), (1, z), (3, y)\}$

## Composition of relations:

Let $A$, $B$ and $C$ be sets, and let $R$ be a relation from $A$ to $B$ and let $S$ be a relation from $B$ to $C$. That is, $R$ is a subset of $A \times B$ and $S$ is a subset of $B \times C$. Then $R$ and $S$ give rise to a relation from $A$ to $C$ denoted by $RoS$ and defined by:

$$a(RoS)c \text{ if for some } b \in B \text{ we have } aRb \text{ and } bSc.$$

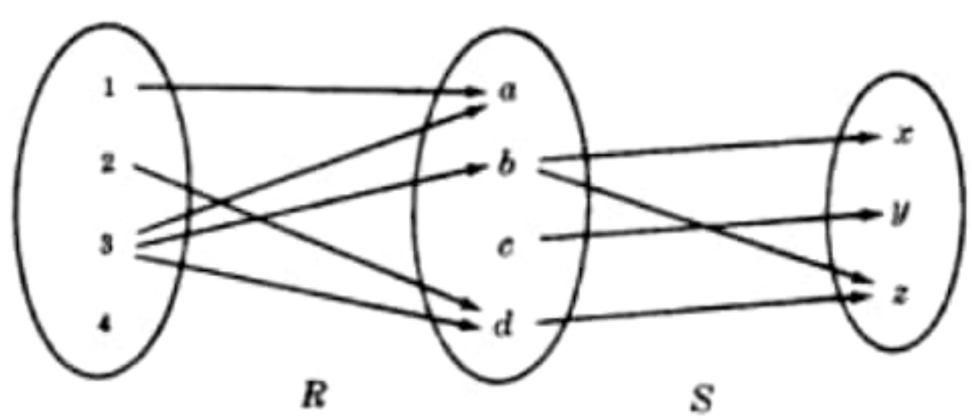
That is,

$$RoS = \{(a, c) \mid \text{there exists } b \in B \text{ for which } (a, b) \in R \text{ and } (b, c) \in S\}$$

The relation $RoS$ is called the composition of $R$ and $S$; it is sometimes denoted simply by $RS$.

Suppose $R$ is a relation on a set $A$, that is, $R$ is a relation from a set $A$ to itself. Then $RoR$, the composition of $R$ with itself, is always defined. Also, $RoR$ is sometimes denoted by $R^2$. Similarly, $R^3 = R^2oR = RoRoR$, and so on. Thus, $R^n$ is defined for all positive $n$.

**Example.** Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$, $C = \{x, y, z\}$ and let $R = \{(1, a), (2, d), (3, a), (3, b), (3, d)\}$ and $S = \{(b, x), (b, z), (c, y), (d, z)\}$ Consider the arrow diagrams of $R$ and $S$ as in following Figure.




Observe that there is an arrow from 2 to $d$ which is followed by an arrow from $d$ to $z$. We can view these two arrows as a "path" which "connects" the element $2 \in A$ to the element $z \in C$. Thus:

$$2(RoS)z \text{ since } 2Rd \text{ and } dSz$$

Similarly, there is a path from 3 to $x$ and a path from 3 to $z$. Hence

$$3(RoS)x \text{ and } 3(RoS)z$$

No other element of $A$ is connected to an element of $C$. Accordingly,

$$RoS = \{(2, z), (3, x), (3, z)\}$$

## Composition of Relations and Matrices:

There is another way of finding $RoS$. Let $M_R$ and $M_S$ denote respectively the matrix representations of the relations R and S. Then

$$M_R = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array} \quad \text{and} \quad M_S = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

Multiplying $M_R$ and $M_S$ we obtain the matrix

$$M = M_R M_S = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \end{array}$$

The nonzero entries in this matrix tell us which elements are related by RoS.

Thus $M = M_R M_S$ and $M_{RoS}$ have the same nonzero entries.

## Types of relations Reflexive Relations:

A relation R on a set A is reflexive if $aRa$ for every $a \in A$, that is, if $(a, a) \in R$ for every $a \in A$. Thus, R is not reflexive if there exists $a \in A$ such that $(a, a) \notin R$.

**Example.** Consider the following five relations on the set $A = \{1, 2, 3, 4\}$:

$R_1 = \{(1, 1), (1, 2), (2, 3), (1, 3), (4, 4)\}$

$R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$

$R_3 = \{(1, 3), (2, 1)\}$

$R_4 = \emptyset$, the empty relation

$R_5 = A \times A$, the universal relation Determine

which of the relations are reflexive.

**Solution:** Since $A$ contains the four elements 1, 2, 3, and 4, a relation $R$ on $A$ is reflexive if it contains the four pairs $(1, 1)$, $(2, 2)$, $(3, 3)$, and $(4, 4)$. Thus, only $R_2$ and the universal relation $R_5 = A \times A$ are reflexive.

Note that $R_1$, $R_3$, and $R_4$ are not reflexive since, for example, $(2, 2)$ does not belong to any of them.

## Symmetric and Antisymmetric Relations:

A relation $R$ on a set $A$ is symmetric if whenever $aRb$ then $bRa$, that is, if whenever $(a, b) \in R$ then $(b, a) \in R$. Thus, $R$ is not symmetric if there exists $a, b \in A$ such that $(a, b) \in R$ but $(b, a) \notin R$.

A relation $R$ on a set $A$ is **antisymmetric** if whenever $aRb$ and $bRa$ then a = b, that is, if $a \neq b$ and $aRb$ then $b \not R a$ .Thus R is not antisymmetric if there exist distinct elements $a$ and $b$ in $A$ such that $aRb$ and $bRa$.

**Example.** Consider the following five relations on the set $A = \{1, 2, 3, 4\}$:

$R_1 = \{(1, 1), (1, 2), (2, 3), (1, 3), (4, 4)\}$

$R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$

$R_3 = \{(1, 3), (2, 1)\}$

$R_4 = \emptyset$, the empty relation

$R_5 = A \times A$, the universal relation

Determine which of the relations are symmetric and antisymmetric.

**Solution:**

**Part-1:**

$R_1$ is not symmetric since $(1, 2) \in R_1$ but $(2, 1) \notin R_1$.

$R_3$ is not symmetric since $(1, 3) \in R_3$ but $(3, 1) \notin R_3$.

The other relations are symmetric.

**Part-2:**

$R_2$ is not antisymmetric since (1,2) and (2,1) belong to $R_2$, but $1 \neq 2$. Similarly, the universal relation $R_3$ is not antisymmetric. All the other relations are antisymmetric.

## Transitive Relations:

A relation $R$ on a set $A$ is transitive if whenever $aRb$ and $bRc$ then $aRc$, that is, if whenever $(a, b), (b, c) \in R$ then $(a, c) \in R$. Thus, R is not transitive if there exist $a, b, c \in R$ such that $(a, b), (b, c) \in R$ but $(a, c) \notin R$.

**Example.** Consider the following five relations on the set $A = \{1, 2, 3, 4\}$:

$R_1 = \{(1, 1), (1, 2), (2, 3), (1, 3), (4, 4)\}$

$R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$

$R_3 = \{(1, 3), (2, 1)\}$

$R_4 = \emptyset$, the empty relation

$R_5 = A \times A$, the universal relation Determine

which of the relations are transitive.

**Solution:**

The relation $R_3$ is not transitive since $(2, 1), (1, 3) \in R_3$ but $(2, 3) \notin R_3$. All the other relations are transitive.

**Example.** Given $A = \{1, 2, 3, 4\}$ and $B = \{x, y, z\}$. Let R be the following relation from $A$ to $B$:

$$R = \{(1, y), (1, z), (3, y), (4, x), (4, z)\} \text{ (a)}$$

Determine the matrix of the relation.

(b) Draw the arrow diagram of $R$.

(c) Find the inverse relation $R^{-1}$ of $R$.

(d) Determine the domain and range of $R$.

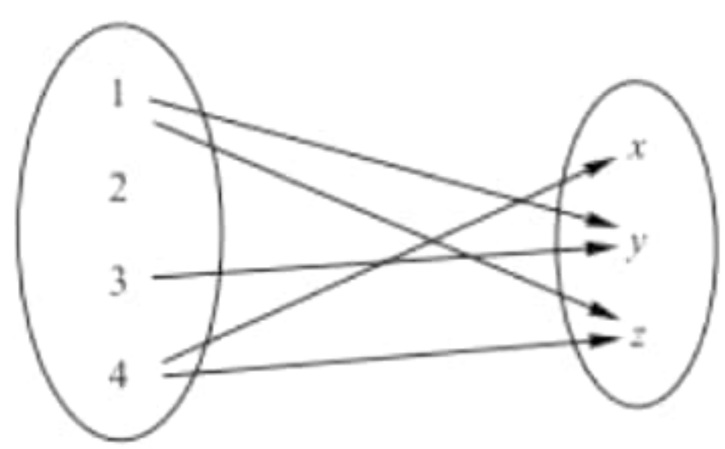
**Solution:**

(a) Here,

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \end{array}$$

From the figure we observe that the rows of the matrix are labeled by the elements of $A$ and the columns by the elements of $B$. Also observe that the entry in the matrix corresponding to $a \in A$ and $b \in B$ is 1 if $a$ is related to $b$ and 0 otherwise.

(b) Here,




From the figure we observe that there is an arrow from $a \in A$ to $b \in B$ iff $a$ is related to $b$, i.e., iff $(a, b) \in R$.

(c) Reverse the ordered pairs of R to obtain $R^{-1}$:

$$R^{-1} = \{(y, 1), (z, 1), (y, 3), (x, 4), (z, 4)\}$$

Observe that by reversing the arrows in the above figure, we obtain the arrow diagram of $R_{-1}$.

(d) The domain of $R$, Dom($R$), consists of the first elements of the ordered pairs of $R$, and the range of $R$, Ran ($R$), consists of the second elements. Thus,

$$\text{Dom}(R) = \{1, 3, 4\} \text{ and } \text{Ran}(R) = \{x, y, z\}$$

**Example.** Let $A = \{1, 2, 3\}$, $B = \{a, b, c\}$, and $C = \{x, y, z\}$. Consider the following relations $R$ and $S$ from $A$ to $B$ and from $B$ to $C$, respectively.

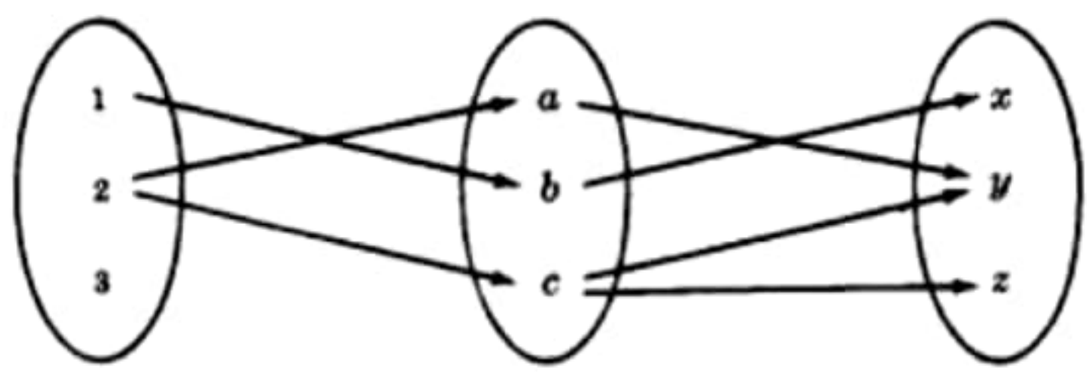
$$R = \{(1, b), (2, a), (2, c)\} \text{ and } S = \{(a, y), (b, x), (c, y), (c, z)\}$$

(a) Find the composition relation $RoS$.

(b) Find the matrices $M_R$, $M_S$, and $M_{RoS}$ of the respective relations $R$, $S$, and $RoS$, and compare $M_{RoS}$ to the product $M_R M_S$.

**Solution:**

(a) Draw the arrow diagram of the relations $R$ and $S$ as in following Figure.




Observe that 1 in $A$ is "connected" to $x$ in $C$ by the path $1 \to b \to x$; hence $(1, x)$ belongs to $RoS$. Similarly, $(2, y)$ and $(2, z)$ belong to $RoS$.

We have

$$RoS = \{(1, x), (2, y), (2, z)\}$$

(b) The matrices of $M_R$, $M_S$, and $M_{RoS}$ follow:

$$M_R = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} a & b & c \end{array} \\ \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{array} \quad M_S = \begin{array}{c} \\ a \\ b \\ c \end{array} \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{array} \quad M_{RoS} = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{array}$$

Multiplying $M_R$ and $M_S$ we obtain

$$M_R M_S = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Observe that $M_{RoS}$ and $M_R M_S$ have the same zero entries.

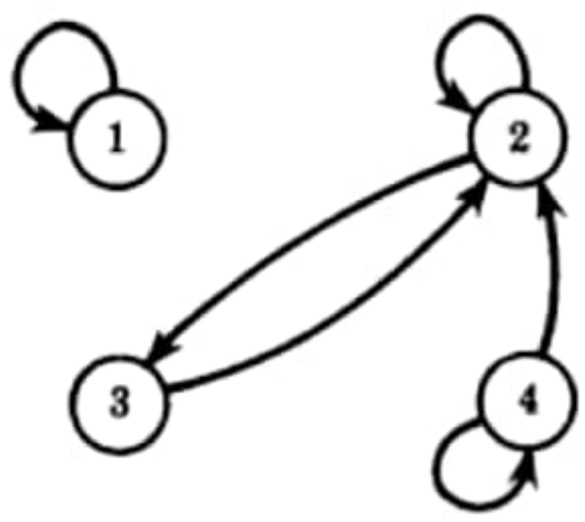
**Example.** Consider the relation $R = \{(1, 1), (2, 2), (2, 3), (3, 2), (4, 2), (4, 4)\}$ on $A = \{1, 2, 3, 4\}$.

(a) Draw its directed graph.

(b) Find $R^2 = RoR$.

**Solution:**

(a) For each $(a, b) \in R$, draw an arrow from $a$ to $b$ as in following Figure.




(b) For each pair $(a, b) \in R$, find all $(b, c) \in R$. Then $(a, c) \in R^2$. Thus

$$R^2 = \{(1, 1), (2, 2), (2, 3), (3, 2), (3, 3), (4, 2), (4, 3), (4, 4)\}$$

**Exercise:**

1. Consider the relation $R = \{(1, 3), (1, 4), (3, 2), (3, 3), (3, 4)\}$ on $A = \{1, 2, 3, 4\}$.

   (a) Find the matrix $M_R$ of $R$.

   (b) Find the domain and range of $R$.

   (c) Find $R^{-1}$.

   (d) Draw the directed graph of $R$.

   (e) Find the composition relation $RoR$.

   (f) Find $RoR^{-1}$ and $R^{-1}oR$.

2. Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$, $C = \{x, y, z\}$. Consider the relations $R$ from $A$ to $B$ and $S$ from $B$ to $C$ as follows:

   $R = \{(1, b), (3, a), (3, b), (4, c)\}$ and $S = \{(a, y), (c, x), (a, z)\}$

   (a) Draw the diagrams of $R$ and $S$.

   (b) Find the matrix of each relation $R$, $S$ (composition) $RoS$.

   (c) Write $R^{-1}$ and the composition $RoS$ as sets of ordered pairs.

3. Let $R$ and $S$ be the following relations on $B = \{a, b, c, d\}$:

   $R = \{(a, a), (a, c), (c, b), (c, d), (d, b)\}$ and $S = \{(b, a), (c, c), (c, d), (d, a)\}$

Find the following composition relations: (a) $RoS$; (b) $SoR$; (c) $RoR$; (d) $SoS$.